

Caio Jordão de Lima Carvalho

SOFTWARE DEVELOPER

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Summary

Current Back-end Developer (Data Engineering) at start-up company Escavador (Potelo Softwares). 3+ years experience specializing in the back-end/data engines development, information retrieval, and machine learning. Super nerd, easy learner, Linux lover and KDE contributor. Problem solver, interested in new challenges, and developing new tools.

Work Experience

Supercomputing Center for Industrial Innovation - SENAI CIMATEC

Salvador, Bahia, Brazil

ARTIFICIAL INTELLIGENCE RESEARCHER

Dec. 2019 - Present

Escavador

Salvador, Bahia, Brazil

BACK-END DEVELOPER

Oct. 2018 - Nov. 2019

- Designed a web crawler architecture for a system focused on information retrieval of Brazilian legal cases, which extracts millions of documents from different courts from Brazil (<https://www.escavador.com/jurisprudencia>).
- Implemented several API microservices using Python, asynchronous libraries and MySQL.
- Project team management.
- Built scripts to extract juridical information from many governmental gazettes.
- Implemented artificial intelligence models focused on information filtering and natural language processing.
- Integrated search engines with support for machine learning ranking using vespa.ai.
- Monitored several web crawlers using Grafana and Prometheus.
- Built fully automated CI pipelines on Gitlab CI for containerized applications using Docker.

KDE Community

Remote

SOFTWARE DEVELOPER & CONTRIBUTOR

Dec. 2017 - Present

- Improved SMART support and KAuth operation flow in KDE Partition Manager during Season of KDE 2018.
- Implemented LVM Volume Group support in Calamares.
- Implemented MDRAID support in KDE Partition Manager during Google Summer of Code 2018.
- Mentored students during Google Code-in 2018, helping them to complete tasks for KDE Partition Manager.
- Mentor representing KDE during Google Code-in 2018 Summit in Google HQ (San Francisco, CA).
- Mentor during Google Summer of Code 2019 in two projects: "Improving Khipu and Analitza" and "Port Authentication to Polkit-qt".
- My patches to KDE Partition Manager and Calamares helped me to improve my skills with C++, C, Qt (widgets), CMake, SMART (libatasmart, smartctl), Dbus, mdadm, dmraid, lvm and Linux partitioning tools.

Labrasoft - IFBA

Salvador, Bahia, Brazil

UNDERGRADUATE RESEARCH ASSISTANT

Aug. 2017 - Aug. 2019

- Developed a simulated stock market to evaluate bots behaviour in decision-making processes using MQL5, MetaTrader and Python.
- Deployed object tracking algorithms based on Deep Learning approaches in embedded systems using Intel Movidius, Raspberry Pi and Python.
- Implemented a generative-adversarial model for image enhancement during Minimally Invasive Surgeries.

TecnoTRENDS

Salvador, Bahia, Brazil

SOFTWARE DEVELOPMENT INTERN

Oct. 2016 - Oct. 2017

- Developer of a module for an educational ERP system focused on elementary schools using C#, ASP.NET and SQL Server.
- Released versions from this module for all the elementary schools from the city of Feira de Santana, Bahia.
- Development of new features, graphical interfaces and SQL routines.

Honors & Awards

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|------|--|----------------------|
| 2018 | Grand Prize Winner , Season of KDE 2018 | Remote |
| 2018 | Winner , Innovative Challenge - IFBA Projects' Hotel | Salvador, Brazil |
| 2017 | Gold Medalist , ProgBASE 2017 - Competitive Programming Contest | C. das Almas, Brazil |
| 2018 | Bronze Medalist , 2nd Programming Contest of Bahia | Salvador, Brazil |

Relevant Presentations

Akademy 2019

Milan, Italy

PRESENTER FOR <KPMCORE IMPLEMENTATIONS AND GOALS FOR THE FUTURE>

September, 2019

- Talked about my work on kpmcore and described some updates and features from KDE Partition Manager 4.0.

QtCon Brazil 2018

São Paulo, Brazil

PRESENTER FOR <COMPUTER VISION WITH QT QUICK AND OPENCV>

Nov. 2018

- Introduced how to integrate the OpenCV library with Qt Quick in mobile environments.

V Workshop Labrasoft-GPETEC

Salvador, Brazil

PRESENTER FOR <INTRODUCTION TO KERAS AND DEEP LEARNING MODELS>

Sep. 2018

- Explained about the Keras library, including an introduction to deep learning models, such as Convolutional Neural Networks, Autoencoders and Generative Adversarial Networks.

ERBASE - Encontro Regional de Computação Bahia-Alagoas-Sergipe

Aracaju, Brazil

PRESENTER FOR <ALPHA B3 - EXPERT ADVISOR USING ARTIFICIAL NEURAL NETWORKS AND GENETIC ALGORITHMS>

Aug. 2018

- Explained about the AlphaB3 project, which is an expert advisor to predict stock market trends based on neural networks and genetic algorithms.

IV Workshop Labrasoft-GPETEC

Salvador, Brazil

PRESENTER FOR <DATA SCIENCE WITH PYTHON>

Aug. 2017

- Introduced some data science topics and explained about some functionalities from Python libraries for data analysis, such as Pandas, NumPy and Scikit-Learn.

Publications

AlphaB3 - Expert Advisor using Neural Networks and Genetic Algorithms to Predict Stock Market Trends

ERBASE 2018

CO-AUTHOR

Aug. 2018

- Paper published in ERBASE 2018. Explained about AlphaB3 and its results on the prediction of the Brazilian stock market.

Organizer

2018 **Organizer**, WebMedia 2018

Salvador, Brazil

2018 **Organizer**, ADS-Integra & ADS-Gênero

Salvador, Brazil

2017 **Problem Writer**, IFBA Competitive Programming Contest

Salvador, Brazil

Education

IFBA (Federal Institute of Science, Education and Technology of Bahia)

Salvador, Brazil

B.TECH. IN SYSTEMS ANALYSIS AND DEVELOPMENT

Jun. 2016 - Oct. 2019

- GPA: 8.6 / 10.0
- Graduation Thesis Title: "Deep Autoencoder Approach to Enhance Low-Light Images during Minimally Invasive Procedures".
- Relevant Coursework: Data Structures, Algorithms, Software Architecture, Design Patterns, Distributed Systems, Software Engineering, Object-Oriented Design, Software Project Management.
- Technical Skills: C++ (3yrs), C, Python, Java, Git, Django, Qt, SQL.
- I was the student representative of this program while studying at IFBA.

College of the Rockies

Cranbrook, Canada

EXCHANGE STUDENT IN APPLIED SCIENCES

Jan. 2019 - May. 2019

- GPA: A-
- Relevant Coursework: Statistics, Probability, Mathematics.
- Selected for a scholarship to the College of the Rockies by the Emerging Leaders in the Americas Program.

SENAI - Serviço Nacional de Aprendizagem Industrial

Salvador, Brazil

CERTIFICATE PROGRAM, COMPUTER NETWORKS AND INFORMATION TECHNOLOGY

Jan. 2015 - Jul. 2016 (Interrupted)

- GPA: 9.5 / 10.0
- Relevant Coursework: Computer Networks, Servers Configuration, Linux and Windows Server, IT Management.

Extracurricular Activity

ProgIFBA - Competitive Programming Group at IFBA

Salvador, Brazil

MEMBER

Dec. 2016 - Jan. 2019

- Created a group focused on competitive programming studies at my university.
- Proposed some training sessions and participated in some contests that have happened in my city.

Sonata Project

MEMBER

Salvador, Brazil

Feb. 2018 - Apr. 2019

- Development of a mobile platform to help people on learning how to read music scores.
- Studied about music information retrieval.
- Worked with a team to submit this project to a innovative challenge at IFBA. Our project was the winner in this competition.